

Antonino Cilione

Data Engineer | Software Engineer

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Summary

Data engineer with 4+ years of experience building distributed data platforms for the energy sector. Develops data pipelines, optimizes Spark workloads, and builds Java/Kotlin backend services. Combines production data engineering with full-stack projects using Python, SQL, TypeScript, and Node.js, including analytical platforms and real-time applications.

Technical Skills

- **Data engineering:** Apache Spark, Kafka, Flink, Iceberg, NiFi, dbt, Airflow, Trino; ETL, stream processing, data quality, data modeling.
- **Programming and backend:** Python, Java, Kotlin, Scala, SQL, TypeScript, JavaScript, C, Bash; Spring Boot, Node.js, Express, React, REST APIs, Socket.IO.
- **Storage and platforms:** PostgreSQL, ClickHouse, MongoDB Atlas, Redis, Parquet, AWS S3, MinIO, Cloudera CDP, Docker, Kubernetes, Git.
- **Software delivery:** Unit and integration testing, GitHub Actions, CI/CD; Prometheus and Grafana in personal projects.

Work Experience

Big Data Engineer - Enel

Jul 2022 - Present

- Build and maintain pipelines for medium- and low-voltage grid measurements using Apache Spark and Kafka, supporting downstream energy-grid analysis.
- Led a migration from Cloudera Distribution Hadoop (CDH) to Cloudera Data Platform (CDP), modernizing the distributed data infrastructure.
- Develop and tune Spark ETL workloads and Apache NiFi ingestion flows; benchmark Apache Iceberg with Spark and Flink.
- Implemented custom equality-delete file commits in Spark workloads using Apache Iceberg.
- Optimize MongoDB Atlas indexes, memory usage, and autoscaling; integrate Redis caching through Redisson in Spark jobs.
- Build Java/Kotlin microservices with Spring Boot to expose MongoDB data through APIs.
- Automate S3 Parquet merge, deduplication, backup, and restore operations with Python, and Kafka/S3 checks with Bash.
- Built a Go desktop tool with Fyne to automate Excel processing, data filtering, and report generation.

Android Developer - Iriscube Reply

Feb 2022 - Jul 2022

- Developed features and fixed defects in a corporate Android banking application.
- Migrated application code from Java to Kotlin and maintained existing functionality.

Education

Bachelor of Engineering in Software Engineering - University of Catania

Oct 2018 - Mar 2022

Thesis: Implementation of learning algorithms on microcontrollers. Developed a Human Activity Recognition (HAR) model on an STM microcontroller using TensorFlow Lite.

Projects

Payment Risk / Pulse

- Built a Java/Flink reference pipeline for synthetic Kafka payments with event-time ordering, deduplication, stateful risk rules, and transactional output; added ClickHouse analytics and a React/TypeScript dashboard for explainable decisions and pipeline health.

Marketplace Lakehouse

- Built a local Kafka/Spark Structured Streaming lakehouse with Iceberg, Trino, SCD Type 2, and data-quality gates; added a TypeScript dashboard, Airflow orchestration, and Prometheus/Grafana monitoring.

Milano Mobility

- Built a Python/PostgreSQL/dbt/MinIO pipeline with versioned archives and tested historical models; added a TypeScript/MapLibre commute explorer and timetable-aware routing API.

Universe Visualizer

- Built a JavaScript/Three.js atlas integrating NASA and Gaia DR3 catalogues, with WebXR virtual and mixed reality, automated tests, and GitHub Actions deployment.

Kamisado

- Built a TypeScript/Node.js/Socket.IO multiplayer game and Electron host with server-authoritative rules, reconnection, and automated rule and socket integration tests.

STM32 Microphone Recorder

- Built C firmware for MEMS microphone capture using DFSDM/DMA and a Python receiver for 16 kHz WAV audio over ST-LINK USB, with buffering, checksums, and host tests.

Interactive Physics Simulations

- Implemented research-based Three.js fluid simulations and Python/Manim animations, with numerical tests for conservation laws, analytical limits, and stopping behavior.

Achievements

- **MALLORN Astronomical Classification Challenge, Kaggle:** Ranked 58th of 894 participants, classifying astronomical transients to identify tidal disruption events.
- **University of Catania Game Jam:** First prize for Greta's Adventures, an environmental platformer developed with Unity and C#.

Certifications and Training

- Improving Deep Neural Networks - DeepLearning.AI, Coursera.
- Algorithm Toolbox - UC San Diego, Coursera.

Languages

Italian: Native. English: Proficient; Cambridge English: First (FCE).